Scrum Artefacts

Product goal, product backlog, sprint log and team retrospective for MARP-Guide.

Snapshot taken on **10 Nov 2026** (Sprint 1 freeze) from the GitHub issues, the *Sprint 1* milestone, pull requests and commit history. The live board is the GitHub Project for this repository.

At freeze: all 32 Sprint 1 issues and all 128 estimated hours are done. The end-to-end run passed for all three members on 8 Nov.

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1. Product Goal

> Build a chat application that answers questions about Lancaster University's Manual of Academic Regulations and Procedures (MARP). Answers come only from the processed MARP PDFs, cite their sources (title, page number, link), and are easy to understand. When no suitable source exists, the answer says "not certain / no source found".

Vision: reliable, quickly accessible information with sources, and no hallucinations.

Users: students and staff who need quick access to university regulations.

Increments:

Increment	Sprint	Scope	Review
1: Core RAG pipeline	Sprint 1, 19 Oct – 10 Nov	Ingestion, extraction, indexing, retrieval and chat as event-driven microservices under Docker Compose; answers with at least 1 citation; Tier 1 and Tier 2 designs	12 Nov 2026
2: Full MVP + features	Sprint 2, 16 Nov – 17 Dec	At least 2 citations, web chat UI, Tier 1 (Auth & Sessions) and Tier 2 (Rate Limiting & Caching) implemented, automated tests in CI, test catalogue	17 Dec 2026

2. Team

Member	GitHub	Sprint 1 track
Amr Khalil Taoumi	AmrTaoumi	A: Ingestion & Extraction
Layla Alwahabi	layla-alw	B: Indexing & Retrieval
Walid Alwahabi	Walid-Alwahabi	C: Chat (LLM, prompt, citations, UI)

Shared items (`track:shared`) are split into one issue per member, e.g. S1-16 (A), (B), (C).

3. Definition of Done

As practised in Sprint 1:

- ☐ Every acceptance criterion in the issue is ticked.
- ☐ Work is done on a `feature/<task-id>-…`, `fix/…` or `docs/…` branch and merged into `develop` through a reviewed pull request.
- ☐ CI is green: `pytest` passes for each service in the matrix and `docker compose build` succeeds.
- ☐ The service starts under `docker compose up` and answers `GET /health`.
- ☐ Any change to an interface also updates `docs/contracts.md`, `docs/events/event-catalogue.md` or the service doc.
- ☐ The issue is closed and the board card is moved to *Done*.

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4. Product Backlog

Every Sprint 1 item is a GitHub issue with owner, estimate, dependencies and acceptance criteria; Sprint 2 candidates become issues at Sprint 2 planning. Sprint 1 estimates are in hours, taken from each issue; Sprint 2 estimates come from the design docs and are in days.

4.1 Sprint 1 items (milestone *Sprint 1*, due 9 Nov 2026)

All 32 issues are done. S1-18 is absent: issues #23–#25 were deleted.

ID	Issue	Item	Track	Assignee	Est.	Status
S1-01	#2	Repo, branches, project board, labels, README skeleton	Shared	Amr	2h	Done 20 Oct
S1-02	#3 #4 #5	Contract workshop	Shared	All three	3h ×3	Done 20 Oct
S1-03	#6	docker-compose with RabbitMQ and Postgres (pgvector) + service template	Shared	Amr	4h	Done 20 Oct
S1-04	#7	Discovery: scrape MARP page, publish `DocumentDiscovered`	A	Amr	6h	Done 26 Oct
S1-05	#8	Fetch and extract PDFs, publish `DocumentExtracted`	A	Amr	8h	Done 2 Nov
S1-06	#9	Ingestion REST API with OpenAPI docs	A	Amr	3h	Done 4 Nov
S1-07	#10	DB schema for chunks + pgvector index	B	Layla	3h	Done 23 Oct
S1-08	#11	Custom chunker with unit tests	B	Layla	8h	Done 24 Oct
S1-09	#12	Embedding and indexing consumer, publish `ChunksIndexed`	B	Layla	6h	Done 4 Nov
S1-10	#13	`/search` endpoint, publish `RetrievalCompleted`	B	Layla	6h	Done 2 Nov
S1-10.1	#46	Retrieval Dockerfile + compose block *(added 2 Nov)*	B	Layla		Done 3 Nov
S1-11	#14	OpenRouter client	C	Walid	4h	Done 2 Nov
S1-12	#15	Prompt template and citation logic	C	Walid	8h	Done 4 Nov
S1-13	#16	Fallback: not certain / no source found	C	Walid	3h	Done 5 Nov

ID	Issue	Item	Track	Assignee	Est.	Status
S1-14	#17	`/chat` endpoint, publish `AnswerGenerated`	C	Walid	6h	Done 8 Nov
S1-15	#18	Minimal static test page for the demo	C	Walid	3h	Done 8 Nov
S1-16	#19 #20 #21	End-to-end run	Shared	All three	8h ×3	Done 8 Nov
S1-17	#22	Event schemas with examples + Mermaid event-flow diagram	Shared	Amr, Layla	3h	Done 31 Oct
S1-19	#26	Tier 1 design doc: User Auth & Sessions	C	Layla	5h	Done 4 Nov
S1-20	#27	Tier 2 design doc: Rate Limiting & Caching (PR #57)	B	Walid	5h	Done 10 Nov
S1-21	#28	Scrum artefacts in `docs/`	Shared	Amr	3h	Done 10 Nov
S1-22	#29	README: architecture diagram, setup, tech stack	Shared	Amr	3h	Done 8 Nov
S1-23	#30 #31 #32	Demo script, slides and reflective report	Shared	All three		Done 10 Nov
S1-24	#33 #34 #35	Freeze: tag `assessment-1`, zip, test on a clean machine	Shared	All three	2h ×3	Done 10 Nov

4.2 Sprint 2 candidates, in priority order

These are refined and turned into issues at Sprint 2 planning (week 6). Rows 1–5 close MVP gaps for Assessment 2. Tier 1 comes before Tier 2 because the per-user rate limit and the cache isolation both depend on the authenticated user (JWT `sub`).

#	Item	Epic	Source	Est.	Owner
1	Answers carry **at least 2 citations** (title, page, link); fall back when fewer than 2 sources qualify	RAG & Chat	Brief 2.1.3 D	TBD	TBD
2	Test catalogue: 10–15 MARP questions with expected answer and source pages	Quality	Brief M9	TBD	TBD
3	Integration tests across services in CI + coverage report	Quality	Brief M8	TBD	TBD
4	Web chat: sources panel with link-out per citation	UX	Brief 2.1.8	TBD	TBD
5	Re-tune `SCORE_THRESHOLD` against the test catalogue and record the scores	Quality	Retro action A2	TBD	TBD
T1-1	Scaffold auth service, compose, `.env.example`, CORS (`CORS_ALLOWED_ORIGINS`)	Tier 1	Tier 1 design §E	0.5d	Amr
T1-2	Create tables on startup (auth: users, refresh_tokens; chat: conversations, messages)	Tier 1	Tier 1 §E	1d	Walid
T1-3	Register and login with hashing and JWT issuing	Tier 1	Tier 1 §E	1.5d	Layla
T1-4	Refresh token rotation and logout revocation	Tier 1	Tier 1 §E	1.5d	Amr
T1-5	Chat: JWT verification, scope `/chat` per user	Tier 1	Tier 1 §E	1d	Walid
T1-6	Chat: conversation list, get and delete endpoints	Tier 1	Tier 1 §E	1d	Layla
T1-7	Publish `UserLoggedIn` / `UserLoggedOut`	Tier 1	Tier 1 §E	0.5d	Amr
T1-8	Web UI: login form, token handling, history view	Tier 1	Tier 1 §E	2d	Walid
T1-9	Tier 1 tests, update `contracts.md` and the architecture diagram	Tier 1	Tier 1 §E	1.5d	Layla
T2-1	Chat DB connection, create tables, env vars, load bge-small in chat	Tier 2	Tier 2 design §E	0.5d	Amr
T2-2	Per-user limit (429) and global LLM budget (503), counting retries	Tier 2	Tier 2 §E	2d	Walid
T2-3	Semantic cache: lookup, save, `cached` field	Tier 2	Tier 2 §E	1.5d	Layla
T2-4	`ChunksIndexed` consumer clears the cache; update the event catalogue	Tier 2	Tier 2 §E	1d	Amr
T2-5	Tune `CACHE_SIMILARITY`; UI messages for 429/503	Tier 2	Tier 2 §E	1d	Walid
T2-6	Tier 2 tests, update `contracts.md` (model, tables) and the diagram	Tier 2	Tier 2 §E	1.5d	Layla

5. Sprint Log

Sprint 1: 19 Oct – 10 Nov 2026

Sprint goal: deliver the core RAG pipeline end to end under Docker Compose (discover → fetch → extract → chunk/embed/index → search → chat with at least 1 citation), with every step publishing an event, and document the Tier 1 and Tier 2 designs.
Milestone: *Sprint 1*, due 9 Nov; freeze and tag `assessment-1` on Tue 10 Nov; upload 11 Nov 12:00; review/presentation 12 Nov.

Sprint backlog

At sprint planning on 19 Oct, 31 issues (S1-01 … S1-24, 128h estimated) were selected into the *Sprint 1* milestone and the sprint board. S1-10.1 (#46) was added on 2 Nov, making 32. The full list with assignees, estimates and status is in 4.1.

Weekly log

Week	Events
19–25 Oct	**Sprint planning** (19 Oct): S1-01 … S1-24 created as issues with owner, estimate, dependencies and acceptance criteria (128h); work split into tracks A/B/C. **Contract workshop** (S1-02, 20 Oct): `contracts.md` agreed before any service code. Compose with RabbitMQ + Postgres/pgvector (S1-03), chunks table with HNSW index (S1-07, PR #37), token-based chunker with tests (S1-08, PR #38). CI pytest workflow added 25 Oct.
26 Oct – 1 Nov	Discovery with robots.txt and dedupe, publishes `DocumentDiscovered` (S1-04, PR #40). **Scope change (29 Oct):** PDF extraction split out of ingestion into its own extraction service; event catalogue written; contracts, service docs and diagram realigned (PRs #41, #42). Event schemas done (S1-17); numbered architecture diagram (PR #43).
2–8 Nov	Extraction (S1-05, PR #44), `/search` with `RetrievalCompleted` (S1-10, PR #45), OpenRouter client (S1-11). Missing retrieval Dockerfile added as S1-10.1. Ingestion API (S1-06), indexing consumer (S1-09, PR #52), citations (S1-12), Tier 1 design (S1-19, PRs #49, #51). Fallback below `SCORE_THRESHOLD` (S1-13), threshold lowered 0.5 → 0.35 after measuring MARP scores (PR #54). 8 Nov: embeddings switched to `bge-small-en-v1.5`, web UI added, **end-to-end run passed** (S1-16).
9–10 Nov	Milestone due 9 Nov with 14h left. **Freeze (10 Nov):** PRs #55–#58 merged; `SCORE_THRESHOLD` raised to 0.6 for bge; Tier 2 design merged (S1-20, PR #57); Scrum artefacts (S1-21), demo prep (S1-23) and freeze (S1-24) done. All 32 issues done.

Sprint 1 outcome

Measure	Result
Issues done	32 of 32 (100%)

Measure	Result
Estimated hours done (velocity)	128 of 128 (100%)
Pull requests merged	21
Services	ingestion, extraction, indexing, retrieval, chat (each with `/health`, in CI and in compose)
Events	`DocumentDiscovered`, `DocumentFetched`, `DocumentExtracted`, `ChunksIndexed`, `RetrievalCompleted`, `AnswerGenerated`
Carried over	None

Sprint 1 review

Increment: a MARP question typed into the web page goes through `/chat` → `/search` over pgvector, then OpenRouter; the answer comes back with numbered citations, or a fallback when no source scores above the threshold. The pipeline is filled by discover → fetch → extract → index, and every step publishes an event on RabbitMQ.
Review: Assessment 1 presentation, 12 Nov 2026.
Feedback: *to be added after the review.*

Sprint 2: 16 Nov – 17 Dec 2026 (planned)

Draft sprint goal: implement Tier 1 (Auth & Sessions) and Tier 2 (Rate Limiting & Caching), and complete the MVP with at least 2 citations, a test catalogue and integration tests in CI.
Planning: week 6, starting from the prioritised list in 4.2 and the actions in 6.3.

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6. Sprint 1 Retrospective

The pipeline shipped on time, but scope changed mid-sprint and core retrieval settings changed in the last days. Two actions carry into Sprint 2.
Format: what went well? / what didn't go well? / what will we improve in Sprint 2? Held at the end of Sprint 1, after the freeze.

6.1 What went well?

- **The contract workshop came first.** S1-02 (20 Oct) produced `contracts.md` and a shared event envelope before any service code. That let the three tracks build in parallel and plug together at the end-to-end run.
- **Issues were well formed.** The Sprint 1 issues created at planning had an owner, dependencies and checkbox acceptance criteria, and most had an hour estimate, so it was always clear what "done" meant.
- **CI came early.** The pytest workflow existed from 25 Oct, and each new service joined the CI matrix as it landed. `docker compose build` also runs in CI.
- **Every change went through a PR into `develop`.** 21 PRs were merged, including a dedicated review PR on the Tier 1 design (#51).
- **We measured before tuning.** The first `SCORE_THRESHOLD` change was based on measured MARP retrieval scores rather than guesswork.
- **The end-to-end run (S1-16) passed for all three members** two days before the freeze.

6.2 What didn't go well?

- **Mid-sprint scope changes.** Splitting extraction into its own service (29 Oct) was the right call, but it forced a pass to realign contracts, service docs and the diagram. The retrieval Dockerfile was missed at planning and added as S1-10.1. The score-threshold fallback was only noticed as missing on 7 Nov.
- **Late changes to core retrieval behaviour.** The embedding model was switched to bge on 8 Nov, and `SCORE_THRESHOLD` moved 0.5 → 0.35 → 0.6 between 7 and 10 Nov. There was no fixed question set to check each change against.

6.3 What will we improve in Sprint 2?

#	Action	Check
A1	Run the end-to-end flow weekly, not just before the freeze	One E2E note per week in the sprint log
A2	Build the test catalogue (backlog #2) **before** further threshold or embedding changes; log scores per change. After any embedding model change, re-index all chunks and re-tune `SCORE_THRESHOLD`	Catalogue committed by week 7